MIL-S-50951(MU)
3 September 1971
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

SWITCH, ROTARY: M21A1

1. SCOPE

1.1 This specification covers one type of a 4-position stepping switch, designated Switch, Rotary: M21A1, which is used to select the desired burst option of the Pershing warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-60902 - Soldering Electrical Connections for Nuclear Weapons Systems, Procedure for.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1169 - Packaging, Packing, and Marking for Shipment of Inert Ammunition Components.

FSC 1135

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DRAWINGS

PICATINNY ARSENAL

8808362	-	Packing and Marking for Storage and Shipments.
9244330	-	Switch, Rotary: M21A1.
9266858	-	Box, Fiberboard, Packing, Ammunition, for Switch, Rotary: M21A1.

PUBLICATIONS

PICATINNY ARSENAL

EL-9244330	-	Inspection Equipment List.
QAI-PA-1	-	Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Materials and components. - Materials shall conform to applicable specifications and drawings. Each component shall conform to all dimensions and tests specified on the component drawing. The contractor shall have available verifiable proof (i.e., objective evidence. See 6.2) that materials and components were fabricated, inspected, and tested under controllable conditions as set forth in the contractor's quality control or inspection procedures.

3.2 Assembly. - The rotary switch shall comply with all requirements specified on drawing (dwg.) 9244330, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawings and herein.

3.3 First article testing.- Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2.1).

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3.4 Coil resistance. - The coil resistance of the switch drive motor shall be measured between pins W and b. The coil resistance shall be as specified in Table I at room temperature and at environmental temperature extremes (when tested during the temperature shock test). In addition, the resistance value (at 77° F.) specified in Table I shall be recorded (see 3.14). Testing shall be as specified in 4.4.1.

TABLE I

Temperature (± 5° F.)	Limits (ohms)	Test Identification Number (see 3.14)*
77° F.	7.7 - 8.9	030001
160° F.	8.9 - 10.7	------
-65° F.	5.1 - 6.4	------

*Not required during temperature shock or vibration testing.

3.5 Contact Resistance. - The contact resistance of the switch shall not exceed 0.050 ohm when measured between the pins specified in Table II. In addition, the maximum contact resistance value (out of a total of three values to be measured for each test point) shall be recorded (see 3.14). Testing shall be as specified in 4.4.2.

TABLE II

(For schematic see dwg. 9244330)

Test Point	From Pin	To Pin	Test Identification Number (see 3.14)
Position 0 (see 6.5)			
1.	S	W	040001
2.	T	W	040002
3.	V	W	040003
4.	C	E	040004
Position 1 (see 6.5)			
5.	U	W	040005
6.	S	W	040006
7.	V	W	040007
8.	B	E	040008
9.	J	X	040009
10.	P	N	040010

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TABLE II (cont.)

(For schematic see dwg. 9244330)

Test Point	From Pin	To Pin	Test Identification Number (see 3.14)
11.	K	Y	040011
12.	R	a	040012
13.	M	a	040013
14.	L	N	040014
Position 2 (see 6.5)			
15.	U	W	040015
16.	T	W	040016
17.	V	W	040017
18.	A	E	040018
19.	G	X	040019
20.	P	N	040020
21.	H	Y	040021
22.	R	a	040022
Position 3 (see 6.5)			
23.	U	W	040023
24.	S	W	040024
25.	T	W	040025
26.	D	E	040026
27.	G	X	040027
28.	H	Y	040028

3.6 Insulation resistance. - The insulation resistance of the switch shall be not less than 50 megohms between pins listed in Table III. The insulation resistance shall be measured after a 10-second application of the test voltage, and the polarity of the test voltage shall be as specified in Table III. There shall be no evidence of insulation breakdown or other damage during this test. Testing shall be as specified in 4.4.3.

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TABLE III

Test Point	From (+)	To (-)
Position 0 (see 6.5)		
1.	A	Case and all other pins
2.	B	"
3.	C	" except E
4.	D	"
5.	G	"
6.	H	"
7.	J	"
8.	K	"
9.	L	"
10.	M	"
11.	N	"
12.	P	"
13.	R	"
14.	S	" except T V W b
15.	U	"
16.	X	"
17.	Y	"
18.	a	"
Position 1 (see 6.5)		
19.	Y	Case and X C D A T H G E S Na
20.	X	" Y C D A T "
21.	C	" Y X D A T "
22.	D	" Y X C A T "
23.	A	" Y X C D T "
24.	T	" Y X C D A "

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TABLE III (cont.)

Test Point	From (+)	To (-)
Position 2 (see 6.5)		
25.	S	Case and a N Y X C B D V M L K J E
26.	a	" S N Y X C B D "
27.	N	" S a Y X C B D "
28.	Y	" S a N X C B D "
29.	X	" S a N Y C B D "
30.	C	" S a N Y X B D "
31.	B	" S a N Y X C D "
32.	D	" S a N Y X C B "
Position 3 (see 6.5)		
33.	V	Case and M R Y L X C B A P Ua K N J E
34.	M	" V R Y L X C B A P "
35.	R	" V M Y L X C B A P "
36.	Y	" V M R L X C B A P "
37.	L	" V M R Y X C B A P "
38.	X	" V M R Y L C B A P "
39.	C	" V M R Y L X B A P "
40.	B	" V M R Y L X C A P "
41.	A	" V M R Y L X C B P "
42.	P	" V M R Y L X C B A "

3.7 Contact load. - The switch contacts, specified in Table II, shall be subjected to a direct current of 10 ± 0.5 ampere for not less than 0.5 second. Upon completion of this test, and after a one-half hour waiting period, the switch shall comply with the requirements of 3.5. Testing shall be as specified in 4.4.4.

3.8 Actuation sequence. - The switch shall be tested to ensure that it will actuate without failure and cycle correctly from position 0 to 1, 0 to 2, and 0 to 3 (in that sequence). The switch shall be monitored for correct cycling to position 0, 1, 2, and 3 by checking continuity between pins listed in Table IV. Testing shall be as specified in 4.4.5.

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TABLE IV

Position (see 6.5)	From Pin	To Pin
0	E	C
1	E	B
2	E	A
3	E	D

3.9 Stepping rate. - The switch shall be capable of being stepped from one position to the next position within a period of 50 milliseconds. The actuation voltage shall be a maximum of 24 volts direct current (VDC), and the stepping time shall be measured from the instant the actuation voltage is applied to the instant the monitor circuit makes contact at the new position. Testing shall be as specified in 4.4.6.

3.10 Contact chatter. - Selected switch contacts, as specified in Table V, shall be monitored for contact chatter during shock and during vibration testing. There shall be no discontinuity or chatter in excess of 10 microseconds. Testing shall be as specified in 4.4.7.

TABLE V

Switch Position (see 6.5)	Series Loop
1	E-B-X-J-N-L-Y-K-a-M
2	E-A-X-G-N-P-Y-H-a-R
3	D-E-G-X-H-Y

3.11 Temperature shock. - The switch shall withstand alternate exposures to temperatures of $160^{\circ} \pm 5^{\circ}$F. and $-65^{\circ} \pm 5^{\circ}$F., for a minimum period of 4 hours at each exposure, and there shall be no evidence of cracking, swelling, distortion, or other damage that may impair its intended operation. While at the 160°F. temperature, the switch shall withstand a relative humidity of $95 \pm 5\%$ without evidence of corrosion or pitting. While at the third cycle of high temperature (at the end of the 4-hour exposure period), the switch shall comply with the requirements of 3.4, 3.5, and 3.9. While at the third cycle of low temperature (at the end of the 4-hour exposure period), the switch shall comply with the requirements of 3.4, 3.5, 3.8, and 3.9. After this test (at room temperature), the switch shall comply with the requirements of 3.4, 3.5, 3.6, and 4.3.3.2. Testing shall be as specified in 4.4.8.

3.12 Impact shock. - The switch shall withstand 18 shocks of 20 ± 1 gravity (g) peak amplitude (sinusoidal) with a shock duration of 12 ± 1 millisecond as measured between points which are 10 per cent of the shock amplitude. The switch shall show no evidence of deterioration or damage that may impair its intended operation. During this test, the

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switch shall comply with the requirements of 3.10. After this test, the switch shall comply with the requirements of 3.5 and 3.8. Testing shall be as specified in 4.4.9.

3.13 Vibration. - The switch shall withstand vibrations as specified in Table VI without evidence of deterioration or damage that may impair its intended operation. During this test, the switch shall comply with the requirements of 3.10. After this test, the switch shall comply with the requirements of 3.4, 3.5, 3.6, 3.8, 3.9, and 4.3.3.2. Testing shall be as specified in 4.4.10.

TABLE VI

Frequency Range cps	Double Displacement or Peak g Level ± 10% Tol.	Minimum Time Period To Cover Freq. Range (minutes)
10 - 30	0.20 inch	2.0
30 - 1000	6 g	2.0
1000 - 2000	6 g	3.5

3.14 Serialization. - A six-letter serial code shall be assigned to each serialized production item. The contractor shall obtain these serial codes by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, Attention: SMUPA-NR2. The letter of request shall include: (a) exact nomenclature of the item as specified on the applicable drawing, (b) contract number or procurement request order number, and (c) number of serial codes required (unused codes must be returned). In addition to the normal inspection records, the contractor shall provide data feedback (see 6.1.2) in conformance with Quality Assurance Instruction QAI-PA-1. The data required to be coded, and the test identification numbers, are specified in the applicable paragraphs in this section.

3.15 Soldering. - All soldering shall be in accordance with MIL-S-60902.

3.16 Workmanship. - All parts and accessories shall be constructed, assembled, and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness, marking, cleaning, and freedom of the parts from burrs and sharp edges which may affect performance or result in injury during handling. All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Submission. - When the contract requires first article inspection, the contractor shall submit 3 consecutively produced rotary switches to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. - The first article inspection samples shall be inspected for the defects specified in 4.3.2.1, using inspection methods contained therein, and for all requirements of Section 3 (also see Table VII), using the test methods specified in 4.4. The rotary switches shall be tested in the order specified in Table VII. If any rotary switch fails to comply with the applicable requirements, the first article samples shall be rejected. Prior to submitting the samples to the testing facility designated (see 4.2.1), the contractor may perform any nondestructive test listed in Table VII (tests (a), (b), (c), (d), (f), and (g)).

TABLE VII

	Test	Requirement Paragraph
(a)	Coil resistance	3.4
(b)	Contact resistance	3.5
(c)	Insulation resistance	3.6
(d)	Hermetic seal	4.3.3.2
(e)	Contact load	3.7
(f)	Actuation sequence	3.8
(g)	Stepping rate	3.9
(h)	Temperature shock	3.11
(i)	Impact shock	3.12
(j)	Vibration	3.13

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4.2.3 Inspection equipment. - The inspection and testing specified in 4.2.2 shall be accomplished with the use of the approved gages and test equipment specified in 4.3.4.

4.3 Quality conformance inspection.

4.3.1 Inspection lot. - A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purposes of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. - A sample shall be selected at random from each inspection lot in accordance with the major defect or the minor defect table as applicable(see Table VIII). Any defect found in the sample, selected for the major or minor classification, shall reject the lot. Critical defects shall require 100 per cent inspection of all items in the inspection lot. If any critical defect is found in an item, the item shall be removed from the lot. The sequence of inspections for the classification of defects need not be followed so long as all the defect inspections are performed.

TABLE VIII

Major Defect Table		Minor Defect Table	
Lot Size	Sample Size	Lot Size	Sample Size
1-30	all	1-30	all
31-45	32	31-45	28
46-60	37	46-60	31
61-75	40	61-75	34
76-95	43	76-95	36
96-130	46	96-130	38
131-180	50	131-180	40
181-250	52	181-250	42
251-400	54	251-400	43
401 and over	56	401 and over	44

4.3.2.1 Classification of defects. - Critical, major, and minor defects are defined in MIL-STD-105.

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4.3.2.1.1 Switch, Rotary: M21A1 (see dwg. 9244330).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Location of mounting hole in case(4 holes)	Gage
	102. Length of case, max	Gage
	103. Width of mounting base, max	Gage
	104. Height, max	Gage
	105. Ability of connector to engage and disengage with mating receptacle	Manual/Gage
	106. Connector keyway improperly located	Visual
	107. Height of pins in connector(see 6.3)	Visual
	108. Connector pin bent, missing, or damaged (26 pins)	Visual
	109. Connector insert material cut or damaged	Visual
	110. Mounting base bent or damaged	Visual
Minor:		
	201. Identification plate missing, improperly marked, or illegible	Visual
	202. Protective finish missing or improper	Visual
	203. Evidence of poor workmanship(see 3.16)	Visual

4.3.3 Inspection testing. - The contractor shall perform all of the tests specified in this paragraph. The tests shall be performed in the following sequence:

	Test	Test Classification
(a)	Coil resistance	Major
(b)	Contact resistance	Major
(c)	Insulation resistance	Major
(d)	Hermetic seal	Major
(e)	Contact load	Major
(f)	Actuation sequence	Major
(g)	Stepping rate	Major
(h)	Impact shock	Major
(i)	Vibration	Major

Of these tests, (e), (h), and (i) are considered destructive and samples so tested shall not be returned to the lot.

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4.3.3.1 Coil resistance (see 3.4), contact resistance (see 3.5), insulation resistance (see 3.6), actuation sequence (see 3.8), and stepping rate (see 3.9). - Each rotary switch shall be subjected to these tests (see applicable paragraph in 4.4). If any switch fails to comply with the applicable requirements, it shall be classified defective and be removed from the lot.

4.3.3.2 Hermetic seal (see dwg. 9244330). - Each rotary switch shall be tested for hermetic seal in accordance with MIL-STD-202, Method 112, Test Condition C, Procedure IV. The reduced pressure of the vacuum chamber, and its duration, shall be compatible with the particular type of Government-approved leak detector used. Perforation of the switch (as specified in Procedure IV) is not required. If any switch fails to comply with the leak rate requirement specified on the drawing, it shall be classified defective and be removed from the lot.

4.3.3.3 Contact load (see 3.7), impact shock(see 3.12), and vibration (see 3.13). - Two rotary switches shall be selected from every lot of 25 switches produced, or portion thereof, and subjected to these tests (see applicable paragraph in 4.4). If any rotary switch fails to comply with the applicable requirements, the lot shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major defects and tests. - Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions as specified on Equipment List, EL-9244330. When either of the following conditions exists, the contractor shall design required equipment or specify suitable commercial equipment: (a) If any deviation from equipment or procedures, as specified on the EL, is desired (b) the code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2). The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions prior to use on the contract(see 6.4).

4.3.4.2 Minor defects and tests. - Not applicable.

4.4 Test methods and procedures. - The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. Unless otherwise specified, the tests shall be conducted at room temperature ($77^{\circ} \pm 10^{\circ}$F.), and the switch shall be set to position 0 (see 6.5).

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4.4.1 Coil resistance. - The coil resistance of the switch drive motor shall be measured with a Government-approved resistance measuring device. The positive lead of the measuring device shall be connected to pin W, and the negative lead shall be connected to pin b. The resistance of the coil shall comply with the requirements of 3.4 during this test.

4.4.2 Contact resistance. - The contact resistance of the switch shall be measured between the pins, and in the sequence, listed in Table II. The contact resistance shall be determined by measuring the voltage drop across the contacts when a direct current of 100 ± 10 milliamperes (ma) is flowing through them. The current shall be supplied by a suitable low-voltage power supply, and the voltage drop shall be measured by a suitable high-impedance voltmeter or electrometer. The contact resistance shall be equal to the voltage drop (in volts) divided by the actual value of current (in amperes) flowing through the contacts. After contact resistance values have been obtained for each of the four switch positions (position 0, 1, 2, and 3), the sequence of readings shall be repeated twice continuing the sequence 0, 1, 2, 3 and 0, 1, 2, 3 (for a total of three readings for each test point). The switch shall comply with the requirements of 3.5 during this test.

4.4.3 Insulation resistance. - The insulation resistance of the switch shall be measured by using a Government- approved insulation resistance tester at an applied voltage of 500 ± 50 VDC. The switch shall comply with the requirements of 3.6 during this test.

4.4.4 Contact load. - The test current shall be applied to the switch contacts through the pins specified in Table II. The test current shall be supplied by a suitable low-voltage power supply. The switch shall not be actuated while current is applied to the contacts. The switch shall comply with the requirements of 3.7 during this test.

4.4.5 Actuation sequence. - The switch shall be actuated to each position by using a maximum of 24 VDC. The switch (initially at position 0) shall be actuated to positions 1, 0, 2, 0, and 3 (in that sequence). While the switch is at each position, a Government-approved continuity tester shall be used to ensure that the switch is correctly positioned by checking continuity between the pins listed in Table IV. The switch shall comply with the requirements of 3.8 during this test.

4.4.6 Stepping rate. - A Government-approved test setup (consisting of a suitable DC power supply and interval timer) shall be connected to the switch to measure the time interval (stepping rate) specified in 3.9.

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The monitor circuit, used to indicate the end of the time interval, shall consist of pins A, B, C, D, and E (see dwg. 9244330 for schematic), and the current through these pins shall not exceed 1 ampere. The switch, initially at position 0, shall be individually stepped to positions 1, 2, 3, and 0 (see 6.5). This sequence shall be repeated twice (for a total of 12 measurements between steps). The switch shall comply with the requirements of 3.9 during this test.

4.4.7 Contact chatter.-Suitable pins of the switch shall be connected together in order to achieve the series continuity loop as specified in Table V (see dwg. 9244330 for schematic). A suitable low-voltage DC power supply shall be used to supply 1.0 +0, -0.2 ampere through each series loop and a Government-approved contact chatter detector. The switch positions during testing are specified in the impact shock and in the vibration test methods (see 4.4.9 and 4.4.10). The switch shall comply with the requirements of 3.10 during this test.

4.4.8 Temperature shock. - The switch shall be placed into a test chamber having an internal temperature of 160°F. and a relative humidity of 95 ± 5%, and it shall remain at this environment for a minimum period of 4 hours. At the end of this test exposure, and within a 5-minute period, the switch shall be transferred to a test chamber having an internal temperature of -65°F. (with uncontrolled humidity), and it shall remain at this temperature for a minimum period of 4 hours. This high- and low-temperature exposure constitutes one cycle, and the switch shall be subjected to three such cycles. The duration of exposure at each extreme temperature shall be not less than that specified and may be extended to overnight exposure to prevent interruption of the transfer sequence. At the end of the third cycle of the high-temperature and low-temperature exposure, the switch shall comply with the applicable requirements specified in 3.11. After completion of testing, at the third cycle of low-temperature, the switch shall be removed from the test chamber, allowed to stabilize at room temperature, and then tested for the applicable requirements specified in 3.11. During this test, the switch shall comply with all requirements of 3.11.

4.4.9 Impact shock. - The switch shall be securely mounted to a shock testing machine, and the switch shall be actuated to position 1 (see 6.5). Applicable switch contacts shall be monitored for contact chatter throughout this test (see 3.10), and the switch shall then be shocked three times along one direction of the longitudinal axis. This process shall be repeated for switch positions 2 and 3 (for a total of 9 shocks). The switch shall then be actuated to position 1 and shocked

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three times in the other direction of the longitudinal axis. This process shall be repeated for switch positions 2 and 3 (for a total of 9 shocks). The switch shall comply with the requirements of 3.12 during this test.

4.4.10 Vibration. - The switch shall be securely mounted to a vibration machine using the same mounting hardware actually used to mount the switch in the missile, and the switch shall be actuated to position 1 (see 6.5). Applicable switch contacts shall be monitored for contact chatter throughout this test (see 3.10). The switch shall be subjected to the vibrations specified in Table VI along the first of three mutually perpendicular axes. This process shall be repeated for switch positions 2 and 3 (along the first axis). The switch shall then be actuated to position1 and vibrated, as specified in Table VI, along the second of three mutually perpendicular axes. This process shall be repeated or switch positions 2 and 3 (along the second axis). The switch shall then be actuated to position 1 and vibrated, as specified in Table VI, along the third of three mutually perpendicular axes. This process shall be repeated for switch position 2 and 3 (along the third axis). The switch shall comply with the requirements of 3.13 during this test.

5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation, packaging, and packing. - Preservation, packaging, and packing shall be level A or C as specified (see 6.1).

5.1.1.1 Level A and level C. - The rotary switch shall be preserved, packaged, and packed in accordance with instructions contained in MIL-STD-1169.

5.1.2 Marking. - Marking shall be in accordance with MIL-STD-1169.

5.2 Spare parts.

5.2.1 Preservation and packaging. - Preservation and packaging shall be level A or C as specified (see 6.1).

5.2.1.1 Level A. - The rotary switch shall be preserved and packaged in accordance with dwg. 9266858,

5.2.1.2 Level C. - The rotary switch shall be preserved and packaged in accordance with dwg.9266858except the barrier material may be omitted.

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5.2.2 Packing.

5.2.2.1 Level A and level C. - The rotary switch shall be packed in accordance with dwg. 8808362.

5.2.3 Marking. - Marking shall be as specified in accordance with dwg. 8808362 and 9266858.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

6.1.1 Procurement requirements:
(a) Title, number, and date of this specification.
(b) Production item or spare part procurement.
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.1.2 Contract data requirements. - Data specified in 3.4, 3.5, and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664, or this specification, when specified on a DD Form 1423 incorporated into the contract.

6.2 Objective evidence. - Records of the contractor's quality control and inspections which can be verified.

6.3 Inspection of connectors.

6.3.1 Height of pins. - In the event a pin does not appear to be the same relative height of adjacent pins, a gage shall be used to determine the variation from normal height. Reject any connector that has a pin that varies by more than 0.040 inch maximum from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface for all measurements.

6.4 Acceptance inspection equipment. - The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, Attention: SMUPA-NR4.

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6.5 Rotary switch positioning. - To set the switch to the desired position, apply 28 ± 4 VDC, unless otherwise specified, to the following pins. Polarity shall be as specified:

Position	From Pin (-)	To Pin (+)
0	b	U
1	b	T
2	b	S
3	b	V

6.6 Supersession data. - This specification includes the requirements of Purchase Description PA-PD-2447 dated 12 May 1964.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project No. : 1135-A121

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